Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: _____ UPF ID: U _ _ _ _ _ _

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

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Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

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Long-answer questions

9. Joint probabilities and dependence (10 points). Let $D = 1$ indicate attendance at an optional career workshop and $Y = 1$ indicate employment after graduation. Their population joint distribution is:

	$D = 0$	$D = 1$
$Y = 0$	0.36	0.24
$Y = 1$	0.04	0.36

   (a) Calculate $\Pr(D = 1)$ and $\Pr(Y = 1)$. (2 points)

   (b) Calculate $\mathrm{E}[Y \mid D = 0]$ and $\mathrm{E}[Y \mid D = 1]$, then use the law of iterated expectations to recover $\mathrm{E}[Y]$. (3 points)

   (c) Calculate $\mathrm{Cov}(D, Y)$ and $\mathrm{Corr}(D, Y)$. (3 points)

   (d) Are D and Y independent? Justify your answer with a probability calculation. (2 points)

10. Comparing two means (10 points). An experiment randomly assigns students to a tutoring group (T) or a control group (C). Treat observations as i.i.d. within each group and independent across groups. The outcome is a test score. The sample summaries are:

Group	Sample size	Sample mean	Sample standard deviation
Tutoring	100	74	12
Control	100	68	16

Let $\Delta = \mu_T - \mu_C$ be the population difference in mean scores.

   (a) Estimate Δ and calculate its estimated standard error. (2 points)

   (b) The claim is that tutoring raises the population mean by exactly two points. State a null and a two-sided alternative, and calculate the test statistic. (2 points)

   (c) Use the supplied critical values to decide whether to reject the null at the 5% and 1% significance levels. Show both comparisons. (2 points)

   (d) Construct a 95% confidence interval for Δ. Does it contain the null value from (b)? Explain how this relates to your test decision. (2 points)

   (e) Plan a new study with m students in each group. Using 12 and 16 as the planning population standard deviations, calculate the smallest integer m for a planned 95% confidence-interval half-width of at most 2 score points. Also report the total planned sample size. (2 points)

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11. A small library population (12 points). The following table describes a population of eight students. G records each student's programme, A or B, and Y is the number of books borrowed last week. The entries are numbers of students, not probabilities. Select one of these eight students uniformly at random.

Books borrowed, Y	Programme A	Programme B
0	1	0
2	2	4
4	1	0

(a) Give the marginal probability distribution of Y and write its complete cumulative distribution function $F_Y(y) = \Pr(Y \le y)$. (3 points)

(b) Calculate $\mathrm{E}[Y]$, $\mathrm{E}[Y^2]$ and the population variance $\mathrm{Var}(Y)$. (3 points)

(c) Calculate $\mathrm{E}[Y \mid G = \mathrm{A}]$ and $\mathrm{E}[Y \mid G = \mathrm{B}]$. Use these conditional means and the programme probabilities to recover $\mathrm{E}[Y]$ by the law of iterated expectations. (3 points)

(d) Are G and Y independent? Justify your answer with a conditional probability calculation. Explain what this example shows about using equal conditional means to infer independence. (3 points)

12. From sample summaries to inference (12 points). An i.i.d. sample of 100 students has assessment scores y_i, measured in points. The following raw sums are available:

$$n = 100, \qquad \sum_{i=1}^{100} y_i = 5200, \qquad \sum_{i=1}^{100} y_i^2 = 280300.$$

Assume that the large-sample standard normal approximation is adequate. Use the two-sided critical values 1.96 at 5% and 2.576 at 1%; no probability-table lookup is needed. You may use the algebraic identity

$$\sum_i (y_i - \bar{y})^2 = \sum_i y_i^2 - n\bar{y}^2.$$

(a) Calculate the sample mean, the centered sum of squares, the unbiased sample variance, the sample standard deviation, and the estimated standard error of the sample mean. State the units of the standard deviation and the standard error. (3 points)

(b) Test the claim that the population mean score equals 50 points against a two-sided alternative. State both hypotheses, calculate the test statistic, and give the decisions at 5% and 1%, showing both critical-value comparisons. (3 points)

(c) Calculate the 95% confidence interval for the population mean. Keep at least two decimal places in its endpoints. Does it contain 50? Relate this answer to your 5% test decision. (3 points)

(d) Plan a new i.i.d. survey using 10 points as the planning population standard deviation. Find the minimum integer sample size for a planned 95% interval half-width of at most 0.98 points. Compare this sample size with 100 and explain the change in precision. (3 points)

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13. Simple OLS: estimation, fit and interpretation (12 points). Using $n = 102$ independent observations, a researcher regresses test score Y_i on weekly independent-study hours X_i, with an intercept: $Y_i = \beta_0 + \beta_1 X_i + U_i$. The observed sample has

$$\bar{x} = 4, \quad \bar{y} = 70, \quad \sum_i (x_i - \bar{x})^2 = 100, \quad \sum_i (x_i - \bar{x})(y_i - \bar{y}) = 50, \quad TSS_Y = \sum_i (y_i - \bar{y})^2 = 425.$$

    (a) Before substituting the sample summaries, write the least-squares objective for the model with an intercept, differentiate with respect to both coefficients, and derive formulas for $\widehat{\beta}_0$ and $\widehat{\beta}_1$. State the condition required for a unique solution. (2 points)

    (b) Calculate $\widehat{\beta}_1$ and $\widehat{\beta}_0$. Write the fitted line and interpret the slope in the units of this example. (2 points)

    (c) For a student with $x_i = 6$ and $y_i = 72$, calculate the fitted score and the residual. (2 points)

    (d) Calculate the explained sum of squares (ESS), the residual sum of squares (SSR), R^2 and SER. (2 points)

    (e) Calculate the sample correlation between study hours and test scores using the supplied centered sums. Verify numerically that the square of this correlation equals R^2. (2 points)

    (f) Study time is now recorded in minutes, $M_i = 60X_i$. Calculate the intercept and slope in the fitted regression of test score on minutes. Report the numerical values of R^2 and SER after this conversion. (2 points)

14. One regressor without an intercept (12 points). Use the same 102 observations and sample summaries given in Question 13. Consider the model $Y_i = \beta_1 X_i + U_i$, with no intercept. Derive the estimator before substituting numbers. Exact fractions are acceptable throughout; decimal division is not required.

    (a) Write the least-squares objective, differentiate it to obtain the first-order condition, and derive a closed-form formula for $\widehat{\beta}_1$. State the denominator condition required for a unique solution. (4 points)

    (b) Calculate $\widehat{\beta}_1$ and interpret the fitted slope in the units of this example. (4 points)

    (c) For the same student with $x_i = 6$ and $y_i = 72$ in Question 13, calculate the fitted score and residual under this no-intercept model. (2 points)

    (d) Use the first-order condition to evaluate $\sum_i x_i \widehat{u}_i$. Calculate the ordinary residual sum $\sum_i \widehat{u}_i$. Must this ordinary sum equal zero without an intercept? Explain. (2 points)

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15. Simple OLS from five observations (12 points). The following table contains five hypothetical students. X is weekly study time in hours and Y is a quiz score in points.

Student i	1	2	3	4	5
x_i	1	2	3	4	5
y_i	4	4	7	8	12

Fit simple OLS with an intercept, $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$. All calculations can be completed by hand; give exact fractions or radicals rather than decimal approximations when appropriate.

(a) Calculate $\bar{x}$, $\bar{y}$, $S_{XX} = \sum_i (x_i - \bar{x})^2$ and $S_{XY} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$. Use them to estimate the slope and intercept. Write the fitted line and interpret the slope. (4 points)

(b) Calculate the fitted score and residual for every student. Verify numerically both $\sum_i \widehat{u}_i = 0$ and $\sum_i x_i \widehat{u}_i = 0$. (3 points)

(c) Calculate $TSS = \sum_i (y_i - \bar{y})^2$, $ESS = \sum_i (\widehat{y}_i - \bar{y})^2$, $SSR = \sum_i \widehat{u}_i^2$, R^2 and SER. Show that $TSS = ESS + SSR$. (3 points)

(d) Study time is instead recorded in minutes, $M_i = 60 X_i$. Calculate the intercept and slope of the same fitted line expressed in minutes. Report R^2 and SER in these units, briefly explaining whether they change. (2 points)